

RESEARCH SUMMARY

Topics	Efficient LLM inference	Discrete Diffusion, Flow Maps, Consistency, Inference-Time Scaling, Speculative Decoding
Publications		G-Star — ICML 2026, poster
Skills	Libraries	PyTorch, Hugging Face Transformers (Qwen, LLaMA)
	Training / inference	Multi-node SLURM (DDP/FSDP), DeepSpeed, FlashAttention, bf16, vLLM/SGLang
	On-device inference	Pruning, quantization, ONNX
Background		5+ years in ML (engineer → researcher), PhD student; available from Jun 2027

PUBLICATIONS

-  **Guided Star-Shaped Masked Diffusion** — V. Meshchaninov, E. Shibaev, [A. Makoian](#), et al. ICML 2026, poster
- Fine-tuning pretrained masked diffusion LMs for guided correction of already generated tokens
-  **Motion ID: Human Authentication Approach** — A. Gavron, et al., [A. Makoian](#) arXiv 2023
- User authentication by motion patterns: $82.2\% \pm 9\%$ accuracy (baseline 70%)

RESEARCH EXPERIENCE

-  **JetBrains** Feb 2025 — Present
ML Researcher / JetBrains Research *with D. Vetrov* Python 3, PyTorch, JAX, transformers, Hydra, Lightning, SLURM
→Masked Diffusion LMs
- Inference-time scaling, diffusion drafters for speculative decoding, curriculum learning (Dream-7B, Qwen3-0.6B MDLM/BD3LM)
- Latent Diffusion Drafters for speculative decoding: lossless acceleration of LLM inference
- Distillation (flow maps, consistency) and latent space compression for real-time acceleration
-  **AIRI (Artificial Intelligence Research Institute)** Jul 2022 — Oct 2022
ML Research Intern / GAN adaptation Python 3, PyTorch, Hydra, WandB
- Few-Shot Image Generation** implementation for few-shot GAN adaptation to a new domain from ~10 images
-  **Yandex** Jul 2021 — Oct 2021
ML Researcher / VR Team *with V. Lempitsky* Python 3, PyTorch
- Trimap-free human alpha matting for VR scenes: improved **FBA Matting**, built alpha-mask datasets
-  Earlier: **Samsung R&D Department**, ML Research Intern, published **Motion ID** Nov 2020 — Jun 2021

EDUCATION

-  **PhD in Computer Science (1st year)**, *Constructor University, Bremen, Germany* Sep 2026 — Present
- Teaching Assistant (seminars): Deep Bayesian Models (Spring 2026), Bayesian Methods (Fall 2026)
 - Co-supervising 3 MSc theses: Loop Transformers, Gaussian Diffusion LMs, Jacobi Decoding
-  **Master in Computer Science**, *Constructor University, Bremen, Germany* Sep 2024 — May 2026
-  **Bachelor in Computer Science**, *Neapolis University of Paphos, Cyprus*, GPA 86/100 Jan 2023 — Jun 2024
-  **Bachelor in Computer Science**, *HSE University, Moscow, Russia*, GPA 8.7/10 Sep 2019 — Jun 2024
-  **Master's Equivalent in Data Science**, *Yandex School of Data Analysis, Russia* Sep 2020 — May 2022

INDUSTRY EXPERIENCE

-  **Yandex** Jun 2023 — Sep 2024
Middle ML Engineer / Geo services Python 3, PyTorch, Hydra, OpenCV, LangChain
- Road segmentation from aerial imagery with IoU 0.93; post-processing reduced artifacts from 10% → 0.02%
 - News classification pipeline with LangChain agents for map update hypotheses: precision 0.9, recall 0.85
-  **Sber Devices** Jul 2022 — Jun 2023
Middle ML Engineer / Computer Vision Platform Python 3, C++, PyTorch, ONNX, torch_pruning
- Real-time background removal for video conferencing (0.98 Dice); pruning gave 1.5x speed-up
 - On-device 700 KB payment card segmentation model with C++ inference in a multiplatform mobile SDK

COMPETITIONS

-  Candidate for the national team at the **IOI** (International Olympiad in Informatics) May, Nov 2018
-  Winner of the national olympiad in **Mathematics** (8th) and **Informatics** (33rd) out of 1M participants Apr 2018, 2019
-  **Top 8%** in Google Hash Code. **Top 50** in Google Kick Start. **Top 350 (top 0.8%)** in Google Code Jam Feb, May, June 2020